between `O(n2)` and `O(n*log(n))`, assuming that the range queries take each `O(log(n))`[17]: just as a reminder, with one million points in our dataset, this means going down from `~1012` (a trillion) operations to `~6*106` (six millions).

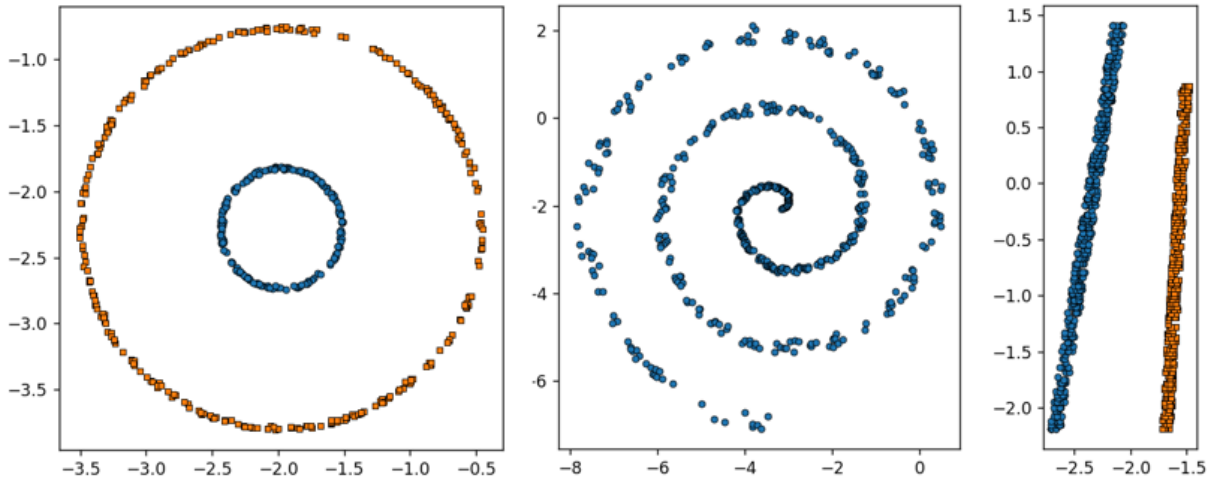


Figure 12.12 The three example datasets in figure 12.6, where k-means was failing: processing them with DBSCAN, and with the right choice of parameters, we can obtain a proper clustering for each of them.

12.3.4 PROs and CONs With DBSCAN

We have already mentioned a few characteristics peculiar to DBSCAN that help overcoming some limits of other clustering algorithms like k-means or single-linkage clustering; let’s quickly review them here:

- DBSCAN is able to determine the number of clusters (given the hyper-parameters with which is called), while k-means needs this number to be provided as a parameter;